

Derivation of the Black-Scholes Formula

Ma Lumping, Giorgia Arcari, Sardor Djamolov,
Federico Di Marcantonio, Tiziano Lostuzzi, Sultonek Eshmirzaev

1 Introduction

We derive the Black–Scholes pricing formula for a European call option using two approaches:

1. the *self-financing replicating portfolio* argument, which yields the Black–Scholes PDE;
2. the *risk-neutral pricing* approach, which gives the formula directly as a discounted expectation under the risk-neutral measure $\mathbb{Q}$.

2 Market Model

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a complete probability space equipped with a filtration $\{\mathcal{F}_t\}_{t \in [0, T]}$ satisfying the usual conditions, and let $W = \{W_t\}_{t \in [0, T]}$ be a standard Brownian motion adapted to $\{\mathcal{F}_t\}$.

We consider a market consisting of two assets:

Definition 2.1 (Risk-free asset). The bond $B = \{B_t\}_{t \in [0, T]}$ evolves deterministically as

$$dB_t = rB_t dt, \quad B_0 = 1, \quad (1)$$

where $r \geq 0$ is the constant risk-free interest rate. Its explicit solution is $B_t = e^{rt}$.

Definition 2.2 (Risky asset). The stock price $S = \{S_t\}_{t \in [0, T]}$ follows a *Geometric Brownian Motion* (GBM), i.e. it satisfies the SDE

$$dS_t = \mu S_t dt + \sigma S_t dW_t, \quad S_0 > 0, \quad (2)$$

where μ is the drift and $\sigma > 0$ is the volatility.

Proposition 2.3 (Explicit solution of the GBM). *By Itô's lemma applied to $f(t, S_t) = \log S_t$, the unique strong solution of the GBM is*

$$S_t = S_0 \exp \left(\left(\mu - \frac{\sigma^2}{2} \right) t + \sigma W_t \right). \quad (3)$$

Remark 2.4. The term $-\frac{\sigma^2}{2}$ is the Itô correction and arises from the quadratic variation $dW_t^2 = dt$.

3 The Self-Financing Portfolio and the Black–Scholes PDE

3.1 The Replicating Portfolio

Let us now consider a self-financing trading strategy where at each time t we hold x_t units of the stock and y_t units of the bond. Let $V = \{V_t\}_{t \in [0, T]}$ be the value of a portfolio holding x_t units of the stock and y_t units of the bond.

Definition 3.1 (Self-financing portfolio). The portfolio (x_t, y_t) is *self-financing* if no money is added or withdrawn.

$$V_t(x_t, y_t) = x_t S_t + y_t B_t. \quad (4)$$

Remark 3.2. In particular, any gains or losses on the portfolio are due entirely to gains or losses in the underlying assets, i.e. the bond and the stock.

We will choose x_t and y_t in such a way that the portfolio (x_t, y_t) replicates the value of the option.

$$\begin{aligned} dV_t(x_t, y_t) &= x_t dS_t + y_t dB_t \\ &= x_t(\mu S_t dt + \sigma S_t dW_t) + r y_t B_t dt \end{aligned} \quad (5)$$

$$= (x_t \mu S_t + r y_t B_t) dt + x_t \sigma S_t dW_t \quad (6)$$

3.2 Deriving the PDE

Let $v(S_t, t)$ be the price at time t of a European option with payoff $v(S_T, T)$. Assume $v \in \mathcal{C}^{1,2}([0, T] \times \mathbb{R}_{>0})$. By Itô's lemma:

$$dv(S_t, t) = \left(\frac{\partial v}{\partial t} + \mu S_t \frac{\partial v}{\partial S_t} + \frac{1}{2} \sigma^2 S_t^2 \frac{\partial^2 v}{\partial S_t^2} \right) dt + \sigma S_t \frac{\partial v}{\partial S_t} dW_t. \quad (7)$$

Set $x_t = \Delta_t = \frac{\partial v}{\partial S_t}$ (delta hedging). The self-financing condition gives:

$$\begin{aligned} dV_t(x_t, y_t) &= \frac{\partial v}{\partial S_t} dS_t + y_t dB_t \\ &= \frac{\partial v}{\partial S_t} (\mu S_t dt + \sigma S_t dW_t) + y_t r B_t dt. \end{aligned} \quad (8)$$

3.3 Elimination of the Stochastic Term

We can equate $dV_t = dv$ and match the dW_t terms: the stochastic parts cancel by construction of Δ_t . Matching the dt terms and using $y_t B_t = V_t - \Delta_t S_t = v - S_t \frac{\partial v}{\partial S_t}$, we obtain:

Theorem 3.3 (Black–Scholes PDE). *Under the above assumptions, the option price $v(S_t, t)$ satisfies*

$$\frac{\partial v}{\partial t} + \frac{1}{2} \sigma^2 S_t^2 \frac{\partial^2 v}{\partial S_t^2} + r S_t \frac{\partial v}{\partial S_t} - r v(S_t, t) = 0 \quad (9)$$

Remark 3.4. Notice that μ has disappeared from the PDE: the option price does not depend on the real-world drift of the stock, only on r , σ , S , K , t and T .

Definition 3.5 (Terminal and boundary conditions). In order to specify the price of which security we are determining boundary conditions must also be provided. For a call option C :

$$C(S, T) = \max(S - K, 0), \quad C(0, t) = 0 \text{ for all } t, \quad C(S, t) \rightarrow S \text{ as } S \rightarrow \infty$$

The solution of the equation in the case of a call option is

$$C(S, t) = S_t \Phi(d_1) - e^{-r(T-t)} K \Phi(d_2)$$

where

$$d_1 = \frac{\log\left(\frac{S_t}{K}\right) + \left(r + \frac{\sigma^2}{2}\right)(T-t)}{\sigma\sqrt{T-t}}$$

and

$$d_2 = d_1 - \sigma\sqrt{T-t}$$

where Φ is the CDF of the normal distribution

Remark 3.6. The Black-Scholes PDE is deduced that the underlying stock does not pay dividend. Assuming instead that the stock pays dividend at the *continuous dividend rate* q , the change in portfolio becomes $dV_t(x_t, y_t) = dv(S_t, t) + \Delta dS_t + \Delta q S_t dt$ and proceeding in the same way, the Black-Scholes PDE becomes:

$$\frac{\partial v}{\partial t} + \frac{1}{2}\sigma^2 S_t^2 \frac{\partial^2 v}{\partial S_t^2} + (r - q)S_t \frac{\partial v}{\partial S_t} - rv(S_t, t) = 0 \quad (10)$$

4 Risk-Neutral Pricing

4.1 The Market Price of Risk and Girsanov's Theorem

Define the *market price of risk* as

$$\theta := \frac{\mu - r}{\sigma}. \quad (11)$$

Theorem 4.1 (Girsanov). Let $\tilde{W}_t := W_t + \theta t$. Define the Radon–Nikodým derivative

$$\left. \frac{d\mathbb{Q}}{d\mathbb{P}} \right|_{\mathcal{F}_t} = \exp\left(-\theta W_t - \frac{1}{2}\theta^2 t\right). \quad (12)$$

Then, under the measure $\mathbb{Q}$, the process $\tilde{W} = \{\tilde{W}_t\}_{t \in [0, T]}$ is a standard Brownian motion.

4.2 Dynamics under $\mathbb{Q}$

Substituting $W_t = \tilde{W}_t - \theta t$ into the GBM:

$$\begin{aligned} dS_t &= \mu S_t dt + \sigma S_t dW_t \\ &= \mu S_t dt + \sigma S_t (d\tilde{W}_t - \theta dt) \\ &= (\mu - \sigma\theta) S_t dt + \sigma S_t d\tilde{W}_t \\ &= r S_t dt + \sigma S_t d\tilde{W}_t. \end{aligned} \quad (13)$$

Hence under $\mathbb{Q}$ the discounted stock price $\tilde{S}_t := e^{-rt} S_t$ is a martingale, and the explicit solution becomes

$$S_t = S_0 \exp\left(\left(r - \frac{\sigma^2}{2}\right)t + \sigma \tilde{W}_t\right). \quad (14)$$

4.3 Risk-Neutral Pricing Formula

Theorem 4.2 (Risk-neutral pricing). *The arbitrage-free price at time $t = 0$ of a European option with payoff $v(S_T, T)$ is*

$$v(S_0, 0) = e^{-rT} \mathbb{E}^{\mathbb{Q}}[v(S_T, T)] \quad (15)$$

where the expectation is taken under the risk-neutral measure $\mathbb{Q}$.

5 The Black-Scholes Formula

5.1 Computing the Risk-Neutral Expectation

Replacing $v(S_T, T)$ by the terminal condition of a European call option $C(S_T, T) = (S_T - K)^+$, we compute $\mathbb{E}^{\mathbb{Q}}[(S_T - K)^+]$ explicitly. Under $\mathbb{Q}$, we have

$$S_T = S_0 \exp\left(\left(r - \frac{\sigma^2}{2}\right)T + \sigma \tilde{W}_T\right), \quad (16)$$

where $\tilde{W}_T \sim \mathcal{N}(0, T)$ under $\mathbb{Q}$. Setting $Z := \tilde{W}_T / \sqrt{T} \sim \mathcal{N}(0, 1)$, we write

$$S_T = S_0 \exp\left(\left(r - \frac{\sigma^2}{2}\right)T + \sigma \sqrt{T} Z\right). \quad (17)$$

The condition $S_T > K$ becomes

$$\begin{aligned} S_0 \exp\left(\left(r - \frac{\sigma^2}{2}\right)T + \sigma \sqrt{T} Z\right) &> K \\ Z &> \frac{\log(K/S_0) - (r - \frac{\sigma^2}{2})T}{\sigma \sqrt{T}} =: -d_2, \end{aligned} \quad (18)$$

where we define

$$d_2 := \frac{\log(S_0/K) + (r - \frac{\sigma^2}{2})T}{\sigma \sqrt{T}}, \quad d_1 := d_2 + \sigma \sqrt{T}. \quad (19)$$

5.2 Splitting the Expectation

We split the expectation as

$$\mathbb{E}^{\mathbb{Q}}[(S_T - K)^+] = \mathbb{E}^{\mathbb{Q}}[S_T \mathbf{1}_{S_T > K}] - K \mathbb{Q}(S_T > K). \quad (20)$$

Second term. Since $S_T > K \iff Z > -d_2$:

$$\mathbb{Q}(S_T > K) = \mathbb{Q}(Z > -d_2) = \mathcal{N}(d_2), \quad (21)$$

where Φ denotes the standard normal CDF.

First term. We compute

$$\begin{aligned} \mathbb{E}^{\mathbb{Q}}[S_T \mathbf{1}_{S_T > K}] &= \int_{-d_2}^{+\infty} S_0 \exp\left(\left(r - \frac{\sigma^2}{2}\right)T + \sigma \sqrt{T} z\right) \frac{e^{-z^2/2}}{\sqrt{2\pi}} dz \\ &= S_0 e^{rT} \int_{-d_2}^{+\infty} \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{(z - \sigma \sqrt{T})^2}{2}\right) dz \\ &= S_0 e^{rT} \Phi(d_1). \end{aligned} \quad (22)$$

5.3 Formula

Combining the two terms and discounting:

Theorem 5.1 (Black–Scholes formula). *The price at time $t = 0$ of a European call option with strike K , maturity T , on a stock with initial price S_0 , volatility σ , under risk-free rate r , is*

$$C(0, S_0) = S_0 \Phi(d_1) - Ke^{-rT} \Phi(d_2), \quad (23)$$

where

$$d_{1,2} = \frac{\log(S_0/K) + \left(r \pm \frac{\sigma^2}{2}\right) T}{\sigma \sqrt{T}}. \quad (24)$$

Remark 5.2. The term $S_0 \Phi(d_1)$ is the risk-adjusted present value of receiving the stock, while $Ke^{-rT} \Phi(d_2)$ is the present value of paying the strike. $\Phi(d_2)$ is the risk-neutral probability that the option expires in the money.

Remark 5.3. The value of the European put option can be calculated by means of the *put-call parity relationship*. Denoted by $P(S_t, t)$ the put price, it is:

$$P(S_t, t) = -S_t \Phi(-d_1) + e^{-r(T-t)} K \Phi(-d_2) \quad (25)$$

6 The Greeks and Dynamic Hedging

6.1 First-Order and Second-Order Greeks

The *Greeks* measure the sensitivity of the option price $v(t, S_t)$ to changes in the model parameters. We work throughout with a European call under the Black–Scholes model.

Definition 6.1 (Delta). Delta measures the sensitivity of C to the underlying price:

$$\Delta := \frac{\partial C}{\partial S} = \Phi(d_1) \in (0, 1). \quad (26)$$

Interpretation. Delta represents the instantaneous rate of change of C per unit move in S . It lies in $[0, 1]$ for calls and $[-1, 0]$ for puts. An at-the-money option has $\Delta \approx 0.5$, reflecting equal probability of expiring in or out of the money. As $S \rightarrow \infty$ (deep in-the-money call), $\Delta \rightarrow 1$; as $S \rightarrow 0$ (deep out-of-the-money call), $\Delta \rightarrow 0$.

Definition 6.2 (Gamma). Gamma measures the rate of change of Delta with respect to the underlying:

$$\Gamma := \frac{\partial^2 C}{\partial S^2} = \frac{\phi(d_1)}{S_0 \sigma \sqrt{T-t}} > 0. \quad (27)$$

Interpretation. Gamma measures the *convexity* of the option price with respect to S , or equivalently the rate at which Delta changes as the underlying moves. A high $|\Gamma|$ means Delta is unstable and requires frequent rebalancing to maintain a hedge. Gamma is always positive for long vanilla options (both calls and puts), reflecting their convex payoff profile. It is maximised at-the-money and near expiry.

Definition 6.3 (Theta). Theta measures the sensitivity of C to the passage of time:

$$\Theta := \frac{\partial C}{\partial t} = -Ke^{-r(T-t)}(r\Phi(d_2) + \frac{\sigma\phi(d_2)}{2\sqrt{T-t}}) \quad (28)$$

where $\phi := \Phi'$ is the standard normal PDF.

Interpretation. Theta captures *time decay*: for a long option position, $\Theta < 0$, meaning the option loses value as time passes, all else equal. Time decay accelerates as expiry approaches, particularly for at-the-money options.

Definition 6.4 (Rho). Rho measures the sensitivity of C to the risk-free rate:

$$\rho := \frac{\partial C}{\partial r} = K(T-t)e^{-r(T-t)}\Phi(d_2), \text{ usually } > 0. \quad (29)$$

Definition 6.5 (Vega). Vega measures the sensitivity of C to volatility:

$$\mathcal{V} := \frac{\partial C}{\partial \sigma} = S(T-t)\phi(d_1) > 0. \quad (30)$$

Interpretation. Vega measures the sensitivity of the option price to changes in volatility. It is always non-negative for long vanilla options: a rise in σ increases the probability of large moves, benefiting option holders regardless of direction. Vega is maximised for at-the-money options and decays as the option moves deep in or out of the money, or as expiry approaches.

6.2 Dynamic Hedging in Practice

A trading desk manages a book of options by continuously adjusting hedge ratios to neutralize unwanted exposures. We denote the portfolio value by Π and derive the hedging conditions systematically from a Taylor expansion of Π .

6.2.1 Delta hedging

Objective: Delta hedging aims to render a portfolio *instantaneously* insensitive to small movements in the underlying price S , achieving a net portfolio Delta of zero:

$$\Delta_{\text{portfolio}} = 0.$$

Construction: Consider a trader who is long one unit of a call option with price $C(S, t)$. The portfolio

$$\Pi = C - \Delta_{\text{call}} \cdot S$$

is Delta-neutral by construction, since:

$$\frac{\partial \Pi}{\partial S} = \frac{\partial C}{\partial S} - \Delta_{\text{call}} = 0.$$

In practice, the hedge is implemented by shorting Δ_{call} units of the underlying (or equivalently going long $|\Delta_{\text{put}}|$ units for a short put position).

Dynamics and Rebalancing: As S and t evolve, Δ changes continuously. A perfect hedge would require continuous trading, which is the idealisation underlying the Black–Scholes derivation. In discrete time, the hedge is rebalanced at intervals δt , incurring a residual *Gamma P&L*:

$$\delta\Pi \approx \frac{1}{2}\Gamma(\delta S)^2 + \Theta\delta t.$$

This expression highlights that a Delta-only hedge does not eliminate uncertainty: the portfolio remains exposed to Gamma risk (convexity) and Theta decay.

6.2.2 Delta-Gamma hedging

Objective: Delta–Gamma hedging extends the first-order hedge by simultaneously neutralising both Delta and Gamma, achieving:

$$\Delta_{\text{portfolio}} = 0, \quad \Gamma_{\text{portfolio}} = 0.$$

This produces a portfolio that is insensitive to both small and moderately large movements in S , reducing the leading-order residual hedging error.

Construction: Since the underlying asset has $\Gamma_S = 0$ (it is linear in S), trading the underlying alone cannot modify the portfolio’s Gamma. A second *non-linear* instrument—typically another option—must be introduced.

Let the portfolio consist of:

- A position in the target instrument (e.g. long one call with price V , Delta Δ_V , Gamma Γ_V),
- w_1 units of a hedging option with price H , Delta Δ_H , Gamma Γ_H ,
- w_2 units of the underlying asset with price S .

The system of equations to solve is:

$$w_1\Gamma_H + \Gamma_V = 0, \tag{31}$$

$$w_1\Delta_H + w_2 + \Delta_V = 0. \tag{32}$$

From (31):

$$w_1 = -\frac{\Gamma_V}{\Gamma_H}.$$

Substituting into (32):

$$w_2 = -\Delta_V - w_1\Delta_H = -\Delta_V + \frac{\Gamma_V}{\Gamma_H}\Delta_H.$$

Interpretation and Residual Risks: By setting $\Gamma_{\text{portfolio}} = 0$, the second-order term in the Taylor expansion of the portfolio P&L vanishes:

$$\delta\Pi \approx \underbrace{\Delta_{\text{portfolio}}}_{=0}\delta S + \frac{1}{2}\underbrace{\Gamma_{\text{portfolio}}}_{=0}(\delta S)^2 + \frac{1}{6}\frac{\partial^3 V}{\partial S^3}(\delta S)^3 + \dots$$

The leading residual is the *Speed* term (third-order), which is negligible for moderate moves. This makes Delta–Gamma hedging significantly more robust than Delta hedging alone, especially for portfolios with large Gamma exposures (e.g. near-expiry or at-the-money options).

6.2.3 Delta–Gamma–Vega Hedging

A fully hedged book requires simultaneous neutralization of three sources of risk: first-order sensitivity to the underlying (Δ), second-order sensitivity to large moves (Γ), and sensitivity to implied volatility ($\mathcal{V}$). These cannot all be neutralized with a single instrument: the underlying carries only Δ exposure, and a single option cannot independently control both Γ and $\mathcal{V}$ since, under Black–Scholes,

$$\frac{\Gamma}{\mathcal{V}} = \frac{1}{S^2 \sigma(T-t)}, \quad (33)$$

which is strike-independent. Two options with the same maturity therefore have identical $\Gamma/\mathcal{V}$ ratios, making simultaneous neutralization impossible. Non-degeneracy is achieved by selecting hedging instruments C_1, C_2 with different maturities $T_1 \neq T_2$.

Proposition 6.6 (Delta–Gamma–Vega neutral portfolio). *Let Π be a book with Greeks $(\Delta_\Pi, \Gamma_\Pi, \mathcal{V}_\Pi)$. Given two hedging options C_i with Greeks $(\Delta_i, \Gamma_i, \mathcal{V}_i)$, $i = 1, 2$, and a position h_S in the underlying, the neutrality conditions are*

$$w_1 \Gamma_1 + w_2 \Gamma_2 = -\Gamma_\Pi, \quad (34)$$

$$w_1 \mathcal{V}_1 + w_2 \mathcal{V}_2 = -\mathcal{V}_\Pi, \quad (35)$$

$$w_1 \Delta_1 + w_2 \Delta_2 + h_S = -\Delta_\Pi. \quad (36)$$

The first two equations form a 2×2 linear system

$$\begin{pmatrix} \Gamma_1 & \Gamma_2 \\ \mathcal{V}_1 & \mathcal{V}_2 \end{pmatrix} \begin{pmatrix} w_1 \\ w_2 \end{pmatrix} = - \begin{pmatrix} \Gamma_\Pi \\ \mathcal{V}_\Pi \end{pmatrix}, \quad (37)$$

which admits a unique solution provided $\Gamma_1 \mathcal{V}_2 - \Gamma_2 \mathcal{V}_1 \neq 0$. Once (w_1, w_2) are determined, Δ -neutrality is restored via

$$h_S = -\Delta_\Pi - w_1 \Delta_1 - w_2 \Delta_2. \quad (38)$$

Remark 6.7. The hedging procedure follows a strict priority: Γ and $\mathcal{V}$ are neutralized first using options, since the underlying contributes nothing to either. Δ -neutrality is restored last using the underlying. All three hedge ratios must be rebalanced dynamically as S , σ , and t evolve. In practice, exact neutrality is unattainable due to transaction costs, discrete rebalancing, and liquidity constraints; desks therefore target approximate neutrality within prescribed risk limits.

References

- [1] F. Black and M. Scholes, *The Pricing of Options and Corporate Liabilities*, Journal of Political Economy, 81(3):637–654, 1973.
- [2] R. C. Merton, *Theory of Rational Option Pricing*, Bell Journal of Economics and Management Science, 4(1):141–183, 1973.
- [3] J. C. Hull, *Options, Futures, and Other Derivatives*, 10th edition, Pearson, 2018.
- [4] N. N. Taleb, *Dynamic Hedging: Managing Vanilla and Exotic Options*, Wiley, New York, 1997.
- [5] M. Haugh, *The Black-Scholes Model*, Lecture Notes, IEOR E4706: Foundations of Financial Engineering, Columbia University, 2016.